

WHAT CAN WE SAY BEFORE WE KNOW WHY?

Description before Explanation

Observation, regularity, model and “how is it so?” in DP Physics

Years 12–13

A physical regularity can be established before a satisfactory explanation is available.



A **description** states what happens: a pattern, dependence, invariance, trend or measured relation.

A **model** introduces selected entities, quantities and relations with which we represent the phenomenon.

An **explanation** answers a question of the form *How is it so?*: what interaction, mechanism, constraint or deeper relation makes the observed regularity intelligible?

These stages need not arrive together. A law can describe extremely well while leaving a deeper explanatory question open.

Technical vocabulary does not guarantee explanation.

“The wave spreads because of diffraction”, “the nucleus decays because it is radioactive”, and “the planet is warmer because of the greenhouse effect” can merely rename the observation.

Ask instead:

What new physical relation has this sentence introduced?

1 Mechanics: from regularity to dynamical account

1. Falling near the Earth

A motion sensor shows that the velocity of a freely falling object changes approximately linearly with time near the Earth’s surface.

Description: what has actually been established?

Model: what idealisations are being made?

Explanation: how is it so that the velocity changes at this rate?

Deeper question: does $F = ma$ explain why gravity exists?

2. Circular motion

An object moves around a circle at constant speed.

Describe the motion without using the words *force* or *centripetal*.

Now explain how constant speed can coexist with non-zero acceleration.

Why is “centripetal force keeps it moving in a circle” incomplete as an explanation?

2 Fields: describing a region, explaining an interaction

A field is especially useful for this distinction. We can first *map an effect in space*. We then introduce a field quantity to describe the region. A further question remains: what produces the field, and how does another system respond to it?

3. Gravitational field

For a spherical mass, measurements are consistent with

$$g \propto \frac{1}{r^2}.$$

What is descriptive about this statement?

How does

$$g = \frac{GM}{r^2}$$

improve the description?

How does the field concept begin to explain the acceleration of a test mass?

What question remains unanswered by introducing the field?

4. Gravitational potential

Outside a spherical mass,

$$V = -\frac{GM}{r}.$$

Explain the distinction between saying that the potential becomes less negative with increasing r and explaining why a mass accelerates inward.

Use

$$g = -\frac{dV}{dr}$$

to connect a spatial description of potential with the local field.

5. Electric field and potential

A small positive test charge accelerates when placed between oppositely charged parallel plates.

Observation:

Field description:

Potential description:

How-is-it-so explanation:

Show how the following belong to different descriptive levels:

$$E = \frac{F}{q}, \quad E = -\frac{dV}{dx}.$$

6. Magnetic field

A compass rotates when moved around a bar magnet. A moving charge can also be deflected in a magnetic field.

What can be described before any microscopic explanation of magnetism is available?

How does a magnetic field map organise these observations?

For a moving charge,

$$\mathbf{F} = q \mathbf{v} \times \mathbf{B}.$$

What does this equation describe particularly well?

What does it *not* by itself explain?

7. Compare the three fields

	Gravitational	Electric	Magnetic
Source quantity			
Test object			
Observable effect			
Can attract?			
Can repel?			
Potential useful?			
Direction of force			

Which similarities are *descriptive analogies*, and which permit common mathematical structure?

3 Thermal physics: transfer, conduction and equilibrium

8. Conduction

One end of a metal rod is heated. Temperature measurements along the rod change with position and time.

Give a purely macroscopic description.

A steady rod approximately obeys

$$P = \frac{kA\Delta T}{L}.$$

What does this relation describe?

Now give a particle-level account of how energy is transferred through the solid.

Why are the equation and the particle account different kinds of scientific achievement?

9. Thermal equilibrium

Two systems in thermal contact approach a common temperature.

What can be known from thermometers alone?

What does the phrase *thermal equilibrium* add?

Does naming the final state explain the approach to it?

Give a “how is it so?” account at an appropriate DP level.

4 Radiation, albedo and the greenhouse effect

10. Black-body emission and emissivity

For a surface at absolute temperature T ,

$$P = e\sigma AT^4.$$

Identify what each part of this equation allows us to *describe*.

What does emissivity e mean physically?

Does the Stefan–Boltzmann law itself explain why thermal radiation has the spectrum it does?

11. Albedo and planetary temperature

A planet reflects a fraction α of the incident solar radiation.

Explain why

$$\alpha = \frac{\text{reflected power}}{\text{incident power}}$$

is a description of a radiative property rather than a mechanism.

If albedo increases while everything else is initially unchanged, predict what happens to absorbed solar power.

How can radiative balance then be used to explain the direction in which equilibrium temperature changes?

12. A simple planetary balance

A simple model equates absorbed solar power with emitted thermal power:

$$(1 - \alpha) \frac{L}{4\pi d^2} \pi R^2 = 4\pi R^2 e \sigma T^4.$$

Before calculating anything:

1. Which terms describe the incoming radiation?
2. Which describe the planet?
3. Which encode geometry?
4. Which equality expresses the equilibrium condition?

Is the calculated T an observation, a description, a model prediction, or an explanation? Defend your classification.

13. Greenhouse effect

Satellite observations show wavelength-dependent absorption and emission by the atmosphere. Write a description that distinguishes incoming short-wave solar radiation from outgoing long-wave terrestrial radiation.

Why is “greenhouse gases trap heat” an unsatisfactory explanation?

Give a better DP-level “how is it so?” account using absorption, re-emission and radiative balance.

What deeper questions remain beyond this account?

5 Waves: laws can precede mechanisms

14. Refraction

A wave crosses from one medium to another. Its frequency remains constant while speed and wavelength change.

Which parts are directly descriptive?

How does

$$v = f\lambda$$

constrain the description without supplying a microscopic mechanism?

What further account would be needed to explain why the wave speed differs between media?

15. Interference

Two coherent sources produce a stable pattern of maxima and minima.

Describe the observed pattern.

Use superposition and phase difference to explain how the pattern arises.

Is “destructive interference makes the wave disappear” a description or an explanation? Repair it.

6 Nuclear physics: statistical description and explanatory model

16. Radioactive decay

Measurements show

$$N = N_0 e^{-\lambda t}.$$

What does this equation describe?

How does the statement “each undecayed nucleus has a constant probability of decay per unit time” explain the exponential form?

Does this explanation tell us when one particular nucleus will decay?

Why is “the nucleus decays because it becomes old” incompatible with the model?

17. Binding energy

A bound nucleus has less mass than the sum of the masses of its separated nucleons.
Give the descriptive statement.

Use

$$E_b = \Delta mc^2$$

to quantify the relation.

Why is “mass is converted into energy” potentially misleading as an explanation?

Give a more careful account in terms of the energy of the bound system and the energy released when it forms.

7 Quantum physics: prediction, representation and explanation

18. Photoelectric effect

Experiments establish that below a threshold frequency no photoelectrons are emitted, however intense the light.

State the observation without explaining it.

Which classical wave expectation does this challenge?

Use

$$hf = \phi + K_{\max}$$

to give the photon-model account.

What has the photon model explained that the equation alone merely relates?

19. Atomic spectra

A low-pressure gas produces discrete spectral lines.

Description:

Model statement:

How-is-it-so explanation using quantised energy differences:

Why would “the atom emits its characteristic colours” be too close to mere description?

20. Wavefunction and probability

A quantum model predicts a probability distribution for repeated measurements.

Separate these statements:

1. what is observed in individual trials;
2. what regularity appears after many trials;
3. what the wavefunction allows us to calculate;
4. what remains an interpretive question.

Why should a successful predictive mathematical object not automatically be identified with a directly observed physical object?

8 Description, explanation, model, or relabelling?

Use

D = description, M = model, E = explanation, R = mere relabelling.

1. The electric field decreases approximately as $1/r^2$. _____
2. The force occurs because there is an electric field. _____
3. A test charge experiences $F = qE$ at that location. _____
4. The metal conducts because it is a conductor. _____
5. At steady state, the temperature gradient along the rod is approximately constant. _____
6. Greenhouse gases warm the planet because of the greenhouse effect. _____
7. Increasing albedo decreases the fraction of incident solar power absorbed. _____
8. The activity falls exponentially with time. _____
9. Radioactive nuclei decay because they are unstable. _____
10. A constant decay probability per unit time produces a constant fractional decrease per unit time. _____
11. Spectral lines occur at discrete wavelengths. _____
12. Photons are emitted with energies corresponding to differences between allowed atomic energy levels. _____

9 One phenomenon, four levels

Choose one:

magnetic field

greenhouse effect

radioactive decay

photoelectric effect

Then write four distinct statements.

1. Observation:

2. Quantitative description:

3. Model:

4. “How is it so?” explanation:

5. A deeper question still left open:

10 Final challenge: when does an equation explain?

For each equation, decide whether it primarily *defines*, *describes*, *constrains*, *predicts*, or contributes to an *explanation*. More than one answer may be defensible.

Equation	Role	What more would an explanation require?
$F = ma$		
$g = GM/r^2$		
$E = F/q$		
$P = e\sigma AT^4$		
$P = kA\Delta T/L$		
$N = N_0e^{-\lambda t}$		
$hf = \phi + K_{\max}$		

11 Reflection

1. Give one example where a precise description is possible before a mechanism is known.
2. Give one example where an equation predicts successfully without answering the deepest “why?” question.
3. What is the difference between naming a phenomenon and explaining how it occurs?
4. Why might a physicist deliberately stop at a model that is known not to be fundamental?

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Teacher notes: framing and selected responses

The central distinction

Description is not a failed explanation. A reproducible regularity is already scientific knowledge. Explanation introduces further structure that makes the regularity intelligible. At DP, the useful additional category is the *model*: a representation can predict and organise observations while leaving deeper explanatory questions unresolved.

Fields

Field equations can be used descriptively: they assign a vector or scalar quantity to positions in space and predict the response of suitable test objects. They also participate in explanation by replacing unexplained action at a distance with local field–object relations. This still leaves deeper questions about the origin and ontology of the fields.

Conduction

$P = kA\Delta T/L$ gives a macroscopic relation between rate of energy transfer, geometry, material and temperature difference. A particle-level account supplies a different explanatory layer: microscopic collisions, lattice vibrations and, in metals, mobile electrons provide channels by which energy is transferred through the material.

Albedo, emissivity and greenhouse effect

Albedo and emissivity are measurable descriptive properties. Radiative-balance models combine them with geometry and the Stefan–Boltzmann law to predict equilibrium temperatures. A greenhouse explanation then adds wavelength-selective atmospheric absorption and emission and the resulting change in the effective level from which radiation escapes to space.

Radioactivity

$N = N_0 e^{-\lambda t}$ describes the ensemble behaviour. Constant decay probability per unit time explains why the same fraction is expected to disappear in equal time intervals and hence why the exponential form arises. It does not predict the decay time of a particular nucleus.

Quantum physics

The photoelectric effect is particularly useful because the observational regularities can be cleanly separated from the photon model introduced to account for them. Likewise, a wavefunction can generate accurate probability distributions without the classroom needing to settle every interpretive question about what the wavefunction “really is”.

A recurring teacher prompt:

Could this sentence have been written from the graph or observation alone?

If yes, it is probably descriptive.

Then:

What additional entity, relation, mechanism or constraint has been introduced?

If the answer is “none”, the explanation may simply have renamed the phenomenon.